APPENDIX

The Story of Virtual Particles

Our discussion of quantum field theory in Chapter Twelve would seem amusingly idiosyncratic to most working quantum field theorists. What we cared about was just the vacuum state, the lowest-energy configuration of a set of quantum fields filling space. But that's just one state out of an infinite number. What most physicists care about are all the other states—those that look like particles moving and interacting with one another.

Just as it's natural to speak about "the position of the electron" when we really know better and should speak about the electron's wave function, physicists who understand perfectly well that the world is made of fields tend to talk about particles all the time. They even call themselves "particle physicists" without discernible embarrassment. It's an understandable impulse: particles are what we see, regardless of what's going on beneath the surface.

The good news is, that's okay, as long as we know what we're doing. For many purposes, we can talk *as if* what really exists is a collection of particles traveling through space, bumping into one another, being created and destroyed, and occasionally popping into or out of existence. The behavior of quantum fields can, under the right circumstances, be accurately modeled as the repeated interaction of many particles. That might seem natural when the quantum state describes some fixed number of particle-like field vibrations, far away from one another and blissfully